Such slope failures could be associated with environmental conditions that used to be different (Dewitte et al., 2021). It is indeed known that the climate and the seismicity conditions of the region have changed over the past tens of thousands of years. For example, the region experienced an abrupt shift from drier conditions to more humid conditions around 13,000 BP (Felton et al., 2007; Wassmer et al., 2013). Although they look non active to the naked eye and that buildings and other infrastructures are present in several of them, old deep-seated landslides can be prone to slope deformations (see Step 2 – Susceptibility assessment for details).

Recent deep-seated landslides: The recent deep-seated landslides seem to be generally associated with quarrying/mining activities as attested by the presence of roads in proximity. These landslides clearly show signs of activity. Caution must be taken when assessing the cause and triggers of these landslides as quarrying/mining sites could either play a cause/trigger role (for example slope undercutting and over steepening) or be the consequence of a landslide. In the latter case, a landslide is at the origin of a fresh regolith/rock outcrop that can be exploited as a commercial opportunity.

Shallow landslides: for the shallow landslides; no features were observed from the satellite images in the four sub-catchments over the last 20 years. However, the fact that we were not been able to identify such landslides, does not allow us to conclude with certainty that shallow slope failures did not occur during that period. Shallow landslides can be very small and therefore happen unnoticed at the resolution of the satellite images. Furthermore, their scars can quickly be altered from weathering, erosion, vegetation regrowth and human activities (Malamud et al., 2004; Dewitte et al., 2021). It should be noted therefore that the identification of the shallow landslides from satellite images is always an underestimation.

Nonetheless, what we can conclude, is that clusters of landslides triggered by heavy rainfall events associated with convective systems (often associated with thunderstorms) did not occur in the studied areas over the last 2 decades. Usually, in the Lake Kivu region, such rainfall events are associated with clusters of hundreds to thousands of slope failures occurring over areas of a few (tenths of) km^2 (Monsieurs et al., 2018; Dille et al., 2019; Depicker et al., 2020a, 2020b, Dewitte et al., 2021); i.e. a disturbance of the landscape that cannot be unnoticed when analysing satellite images. On a yearly basis, a few of such events are observed in the Lake Kivu region and their spatial distribution is controlled by the randomness of the extreme convective rainfall events. In Figure 3 a, two clusters of landslides corresponding to such climatic events are observed.

Soil creep: Soil creep is a movement under the influence of gravity, temperature and moisture fluctuations, and the action of biota that occur on hillslopes. It is characterized by an extremely slow, generally imperceptible displacement of surficial unconsolidated materials that can occur over long periods of time, easily over thousands of years (Thomas, 1994; Heimsath and Jungers, 2013). This process is common in many tropical regions, particularly where, like in Rwanda, the regolith is thick, and, although imperceptible, it can displace a considerable volume of hillslope material downstream (Thomas, 1994).